Practice — Energy and Speed Are Related

Chapter 3 · Grade 4

Name: _____ Date: _____

A. SAME OBJECT OR NOT?

Which comparisons are fair (the SAME object at different speeds)? Circle SAME or NOT.

Comparison	Fair (same object)?
One ball rolled slowly, then the same ball rolled faster.	SAME / NOT
A small ball and a big ball rolled at the same speed.	SAME / NOT
One marble dropped from 30 cm, then 60 cm.	SAME / NOT
A heavy cart and a light cart pushed the same way.	SAME / NOT

B. READ THE DATA

A group tested ONE ball down a ramp at three heights.

Ramp Height	Average Cup Displacement	Faster or slower?
1 book	6 cm	
2 books	14 cm	
3 books	22 cm	

Which ramp height made the ball move fastest? _____ How do you know? (You did not measure speed.)

C. READ THE GRAPH

A class graphed Ramp Height vs. Average Cup Displacement. The tallest bar is at 3 books, the shortest at 1 book.

What does this graph show about speed and energy of motion for the SAME ball?

D. CLAIM + EVIDENCE

Which evidence supports the claim: “Faster motion of the same object is associated with greater energy of motion”? Circle all that fit.

1. The same ball moved the cup farther from a higher ramp.
2. The ball is blue.
3. The same ball made a bigger crater from a higher drop.
4. A heavier ball moved the cup farther.

E. FIX THE SCIENTIST

A scientist says: “The heavier ball had more energy, so speed must be the reason.”
What is wrong with this claim? (Hint: what changed besides speed?)
